Magnetosphere and changing fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

SUCCESS CRITERIA - I can...

1. Connect changing magnetic fields with induced electric effects.
2. Distinguish an assumption from an observation.
3. Name evidence required for a space-weather prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Near an ideal long straight wire, current becomes 3 times as large and distance becomes 5 times as large. Compare B.

Marzano check - circle: ① ② ③

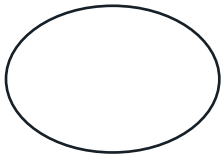
02 Worked example: explain the model

Reference: $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$; $\Phi = BA$.
Uniform perpendicular field for $\Phi = BA$. Use supplied/measured field changes; an asteroid alone is not evidence of an electromagnetic pulse.

Worked reasoning: A changing magnetic environment can induce voltages in conducting loops. A static magnetic field alone is not the same as a changing flux through a particular circuit.

Your turn: Sketch a conducting loop and field arrows; show one way flux can change.

Label field direction, loop area and what changes.



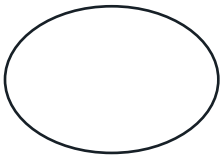
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a coil in a changing magnetic field. Label what changes and where an induced emf could appear.

Label field direction, loop area and what changes.



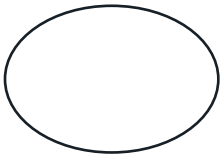
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical 10-turn loop of area 5 m² experiences a uniform perpendicular field change of 2×10^{-6} T over 7 s. Label the given quantities and calculate average emf magnitude. Does this calculation establish what caused the field change?

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A headline claims an approaching rock will disable power grids through an EMP, but gives no field measurements or mechanism. Name two pieces of evidence needed before accepting it.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two series components have drops 4 V and 6 V. A charge of 8 C passes through each. Find each energy transfer and the total.
Label voltage, charge and energy with their distinct units.

Initial / supply _____

V: ____ J/C

Final / component _____

V: ____ J/C

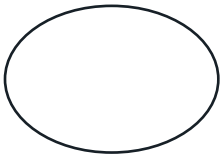
Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

07 Exit check

Explain why a one-loop calculation cannot by itself predict effects on a whole power grid.

Label field direction, loop area and what changes.



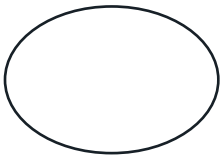
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A one-turn 100 m² loop is perpendicular to a uniform field that changes by 2 microtesla in 0.5 s. Estimate emf. Explain why this toy result is not a full forecast for a power grid.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3